Chapter 4 — Work, Energy and Power

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Topic 1 — Work done by constant and variable forces

Q.1 Explain work done by a constant force and show how work done by a variable force is found from a graph.

Work is done when a force produces a displacement in its own direction. For a constant force, $W = Fd \cos \theta$, where θ is the angle between force and displacement. Work is a scalar measured in joules. It is zero when the displacement is zero or when the force is perpendicular to the displacement. Pushing a wall does no work because the wall does not move.

When the force varies, the displacement is divided into small strips over which the force is nearly constant. The work in each strip is $F \cos \theta \, \Delta d$, and the total work is the sum of these strips. This sum is the area under the graph of $F \cos \theta$ against d. For a spring, the force increases linearly with extension, so the area is a triangle and $W = \frac{1}{2}Fx = \frac{1}{2}kx^2$.

Important Conceptual Short Questions — Topic 1

1. Why is no work done when a force acts perpendicular to the displacement?

Work depends on the component of force along the displacement, $F \cos \theta$. At $\theta = 90°$ this component is zero, so the force changes direction but does not transfer energy along the path.

2. How does a force–displacement graph give work done?

Each narrow strip of area $F \, \Delta d$ represents the work done over a small displacement. Adding all strips gives the total work, so the total area under the graph equals the work done.

3. Why is work a scalar quantity although force and displacement are vectors?

Work is the dot product of force and displacement, $F \cdot d = Fd \cos \theta$. A dot product gives a scalar, so work has magnitude only and no direction.

Topic 2 — Conservative and non-conservative forces

Q.2 Differentiate between conservative and non-conservative forces with examples.

A conservative force is one for which the work done depends only on the initial and final positions, not on the path taken. The work done around a closed path is zero. Gravity, electrostatic force and spring force are conservative. A non-conservative force is one for which the work depends on the path. Friction is the common example: the longer the path, the more work is done against friction, and the work around a closed path is not zero.

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Feature	Conservative force	Non-conservative force
Path dependence	Work depends only on endpoints	Work depends on the path
Closed path	Total work is zero	Total work is not zero
Potential energy	Can be defined	Cannot be defined
Examples	Gravity, electrostatic force, spring force	Friction, air resistance

Important Conceptual Short Questions — Topic 2

1. Why is the work done by gravity around a closed path zero?

Gravity is conservative. The work done in moving up is negative and equal in magnitude to the positive work done in moving down, so the total work over a closed path cancels to zero.

2. Why is friction called a non-conservative force?

The work done against friction depends on the length of the path. A longer path produces more heat, and the work around a closed path is not zero, so friction is non-conservative.

3. How does a conservative force allow potential energy to be defined?

Because the work depends only on position, a unique potential energy can be assigned to each point. The change in potential energy equals the negative of the work done by the conservative force.

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Topic 3 — Power and energy units

Q.3 Define power and explain the kilowatt-hour as a unit of energy.

Power is the rate of doing work or transferring energy: $P = W/t$. For a constant force moving a body with velocity v, $P = F \cdot v$. The SI unit is the watt, where $1\ W = 1\ J\ s^{-1}$. A machine of high power does the same work in less time.

The kilowatt-hour is a commercial unit of energy, not power. It is the energy used by a device of power 1 kW operating for 1 hour: $1\ kWh = 1000\ W \times 3600\ s = 3.6 \times 10^6\ J = 3.6\ MJ$. Electricity bills are charged in kWh because it measures the total energy consumed.

Important Conceptual Short Questions — Topic 3

1. Why is the kilowatt-hour a unit of energy and not power?

A kilowatt-hour is power multiplied by time, and power × time = energy. It measures the total energy used in one hour at a power of one kilowatt.

2. How can the same work be done with different power?

Power is work divided by time. Doing the same work in a shorter time requires greater power, while a longer time requires less power.

3. Why is P = F·v useful for moving vehicles?

It shows that for a given engine force, power increases with speed. It also shows that at constant power, a larger driving force is available at lower speed, which is why vehicles use low gears for climbing.

Topic 4 — Kinetic energy and the work–energy theorem

Q.4 Derive the expression for kinetic energy using the equations of motion and state the work–energy theorem.

A body of mass m initially at rest is acted on by a constant force F and moves a distance d, reaching speed v. From $v^2 = u^2 + 2ad$ with $u = 0$, $a = v^2/2d$. The work done is $W = Fd = mad = m(v^2/2d)d = \frac{1}{2}mv^2$. This work is stored as kinetic energy, so $K = \frac{1}{2}mv^2$.

The work–energy theorem states that the net work done on a body equals the change in its kinetic energy: $W_{net} = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$. It applies whether the force is constant or variable and is especially useful when time is not involved.

Important Conceptual Short Questions — Topic 4

1. Why does kinetic energy depend on the square of speed?

The work needed to accelerate a body from rest is proportional to v^2 because the distance required under constant acceleration increases as v^2. Doubling the speed therefore requires four times the kinetic energy.

2. How does the work–energy theorem simplify problems without time?

It relates net work directly to the change in kinetic energy, so problems involving force, distance and speed can be solved without finding acceleration or time.

3. Why can kinetic energy never be negative?

Kinetic energy is $\frac{1}{2}mv^2$. Mass is positive and v^2 is always positive or zero, so kinetic energy is always positive or zero.

Topic 5 — Gravitational potential energy and escape velocity

Q.5 Derive the expression for absolute gravitational potential energy and use it to find escape velocity.

Near the Earth’s surface, potential energy is $U = mgh$, but this assumes constant g. For large distances the gravitational force varies, so the absolute potential energy is defined with zero at infinity. The work done by gravity in bringing a mass m from infinity to distance R from the centre of the Earth is negative, giving $U = -GMm/R$. The negative sign shows that the field is attractive and that work must be done to remove the mass to infinity.

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Escape velocity is the minimum speed needed to escape the gravitational field without further propulsion. Setting kinetic energy equal to the magnitude of potential energy, $\frac{1}{2}mv_e^2 = GMm/R$, gives $v_e = \sqrt{(2GM/R)} = \sqrt{(2gR)}$. For the Earth this is about 11 km s^{-1}. It is independent of the mass of the escaping body.

Important Conceptual Short Questions — Topic 5

1. Why can mgh not be used for a rocket leaving the Earth?

mgh assumes that g is constant, which is true only near the surface. Over large distances g decreases with height, so the absolute potential energy expression $-GMm/R$ must be used.

2. Why is gravitational potential energy negative?

Zero potential energy is chosen at infinity. Since gravity is attractive, work must be done against the field to move a mass to infinity, so the potential energy at any finite distance is less than zero.

3. Why is escape velocity independent of the mass of the body?

Both kinetic energy and gravitational potential energy are proportional to the mass of the body. The mass cancels when $\frac{1}{2}mv^2 = GMm/R$ is solved for v.

Topic 6 — Interconversion of energy and conservation

Q.6 Explain the interconversion of potential and kinetic energy and the effect of friction on conservation.

In the absence of friction, the total mechanical energy K + U remains constant. A falling body loses potential energy and gains an equal amount of kinetic energy; a pendulum or swing continuously changes energy between the two forms. At the highest point the energy is mostly potential, and at the lowest point it is mostly kinetic.

When friction is present, mechanical energy is not conserved because some energy becomes heat and sound. The total energy of the system and surroundings is still conserved. For a body falling through a resistive medium, the book writes $mgh = \frac{1}{2}mv^2 + fh$, where fh is the work done against the resistive force.

Important Conceptual Short Questions — Topic 6

1. Why does a pendulum eventually stop although energy is conserved?

Friction and air resistance convert mechanical energy into heat and sound. The total energy is conserved, but the mechanical energy of the pendulum decreases until it stops.

2. How does a swing show interconversion of energy?

At the highest point the swing has maximum potential energy and minimum kinetic energy. At the lowest point potential energy is minimum and kinetic energy is maximum, so energy changes between the two forms.

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3. Why is total energy conserved even when friction acts?

Friction does not destroy energy; it converts mechanical energy into internal energy of the surfaces and surroundings. The sum of all forms of energy remains constant.

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